

Problem 1.**Airport data:**(a) What is the α if the distribution is power law?

$\alpha = 1.612$, with $x_{\min} = 1$ route. Maximum-likelihood estimate $\hat{\alpha} = 1 + n / \sum_i \ln(x_i / x_{\min})$, $x_{\min} = \min_i x_i$, $n = 3409$.

$n = 3409$	number of airports
$x_{\min} = \min_i x_i = 1$	smallest route count
$S = \sum_{i=1}^{3409} \ln(x_i / 1) = 5569.43$	sum of log ratios
$n / S = 3409 / 5569.43 = 0.6121$	
$\hat{\alpha} = 1 + 0.6121 = 1.612$	estimate

(b) What is the λ if the distribution is exponential?

$\lambda = 0.0504$. MLE $\hat{\lambda} = 1 / \bar{x} = 1 / 19.845$.

$\sum_i x_i = 67652$	total routes
$\bar{x} = 67652 / 3409 = 19.845$	mean
$\hat{\lambda} = 1 / 19.845 = 0.0504$	estimate

(c) What is $[a, b]$ if the distribution is uniform?

$[a, b] = [1, 915]$, the minimum and maximum route counts (MLE).

$\hat{a} = \min_i x_i = 1$	fewest routes
$\hat{b} = \max_i x_i = 915$	most routes

(d) What is μ and σ if the distribution is normal?

$\mu = 19.845$, $\sigma = 53.506$ (sample mean and sample standard deviation with $n - 1$).

$\hat{\mu} = \bar{x} = 19.845$	mean, as in the exponential fit
$\sum_i (x_i - \bar{x})^2 = 9,756,906.22$	sum of squared deviations
$9,756,906.22 / (3409 - 1) = 2862.94$	sample variance
$\hat{\sigma} = \sqrt{2862.94} = 53.506$	standard deviation

(These can be CDF, PDF, or QQplot)

For (e) to (i):

- **Samples.** $n = 3409$ random values drawn from each fitted model (seed 6330).
- **Colours.** Data in blue, model sample in orange.

- *D*. Two-sample Kolmogorov–Smirnov distance: the largest vertical gap between the two CDFs (0 = identical).

(e) Screen shot of the visualization of distribution of data

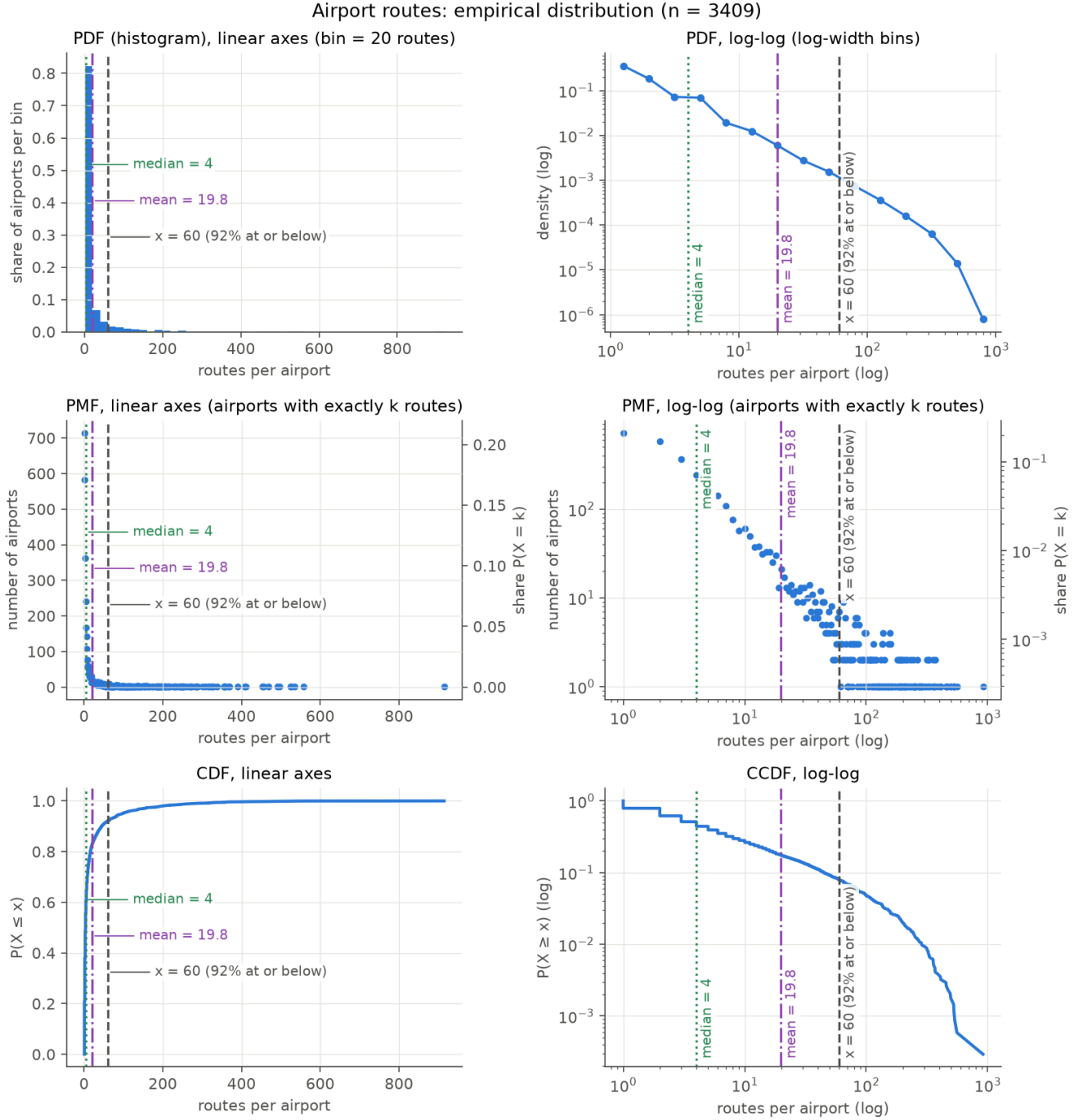


Figure 1: Airport routes: empirical distribution ($n = 3409$). Rows: PDF, PMF, CDF/CCDF. Left column on linear axes, right column on log-log axes.

PDF, linear (top left). Histogram with 20-route bins; each bar is the share of airports in its bin. One tall bar at the left edge: 82% of airports have fewer than 20 routes, and the bars beyond 200 are almost flat on the floor.

PDF, log-log (top right). The same distribution with bins of equal width on the log scale, divided by

bin width so it is a density. It falls along a roughly straight line and bends down above about 100 routes.

PMF, linear (middle left). One dot per route count k : the number of airports with exactly k routes (left axis) and the same height as a share $P(X = k)$ (right axis). 713 airports have 1 route (20.9%), 583 have 2, 363 have 3; above about 30 routes every count is at most 14 airports and lies on the floor.

PMF, log-log (middle right). The same dots on log axes. Up to about 30 routes they fall on a line of slope about -1.45 , close to $-\hat{\alpha}$. Right of the dashed line ($x = 60$) each route count belongs to only 1, 2 or 3 airports, so the dots sit in rows at exactly those counts; that is a counting floor, not the shape of the distribution.

CDF, linear (bottom left). The curve rises almost vertically and reaches 0.92 at 60 routes, then stays flat up to the maximum of 915.

CCDF, log-log (bottom right). Share of airports with at least x routes. It is close to a straight line up to about 100 routes, then bends down and ends at the busiest airport (915 routes).

Reference lines. In every panel the green dotted line is the median (4 routes: half the airports have at most 4), the purple dash-dot line is the mean (19.8 routes), and the grey dashed line is $x = 60$ (92% of airports lie at or left of it). The mean sits five times to the right of the median because a few hubs with hundreds of routes pull it up; on the log-log CCDF only about 18% of airports reach the mean. On linear axes all three lines are squeezed against the left edge.

Summary. The median airport has 4 routes and 38% have only 1 or 2, while the busiest has 915. The mean (19.8) is five times the median and the skewness is 6.0, so a few hubs hold most of the routes.

(f) Screen shot of the visualization of distribution of data if it was power law

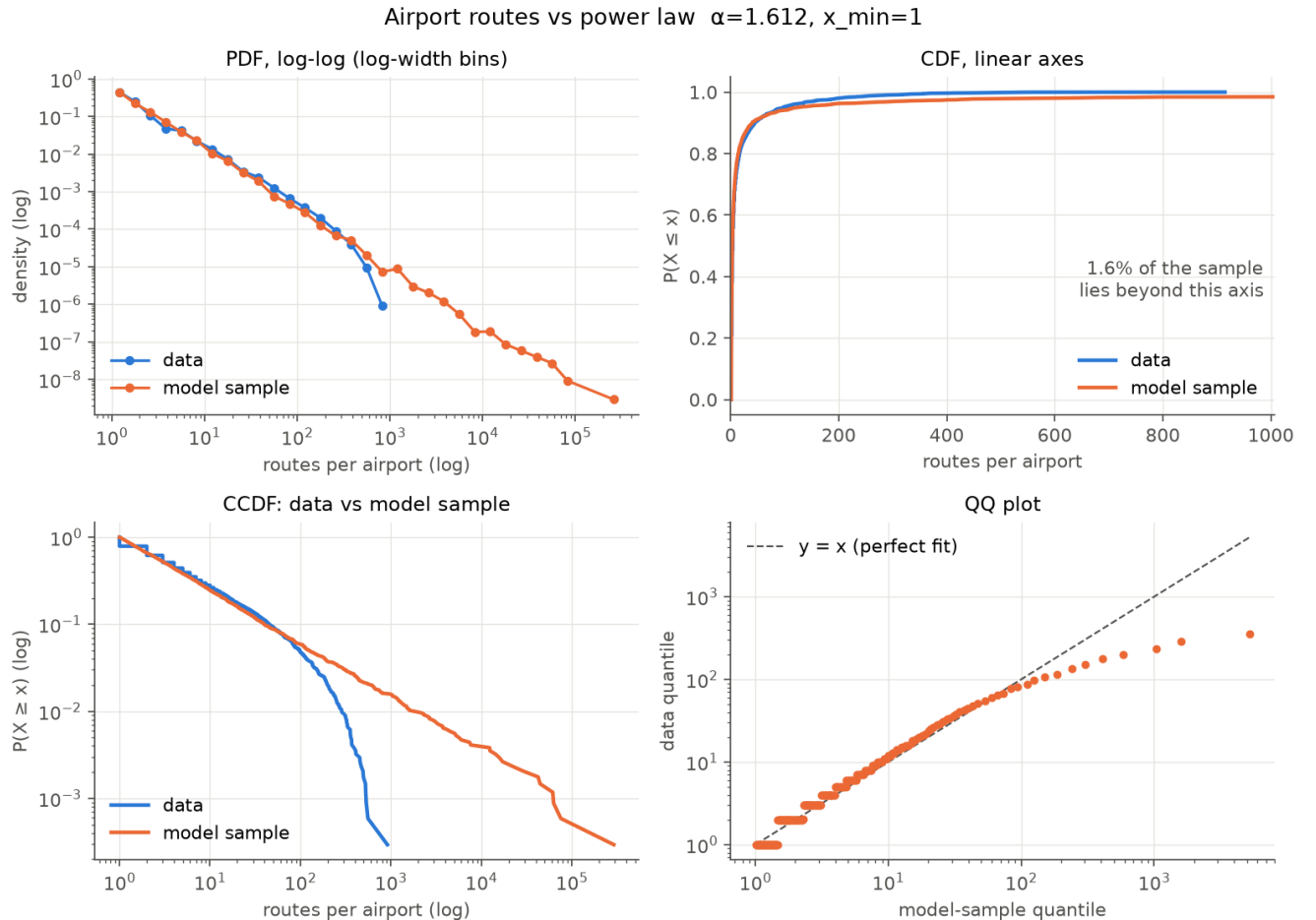


Figure 2: Airport routes vs. power law ($\alpha = 1.612$, $x_{\min} = 1$). $D = 0.209$.

PDF (top left). The two lines follow the same straight slope up to about 300 routes. Above that the data drops and ends near 1000, while the model continues past 10^5 (its largest sampled airport has 291,593 routes).

CDF (top right). On linear axes the two curves almost overlap. The difference is hidden in the 1.6% of the sample that lies beyond 1000 routes.

CCDF (bottom left). Both follow the same straight line on log-log axes up to about 100 routes. Above that the data bends down and stops at 915; the model stays straight.

QQ plot (bottom right). The points lie on $y = x$ up to about 60 routes, then run off to the right, below the line: at the 99.5th percentile the model has 5,234 routes and the data 358.

Why the CCDF on log-log axes. For a power law, the CCDF is the complement of the CDF, and taking the log of both sides turns it into a straight line:

$$\begin{aligned}
 P(X \geq x) &= 1 - P(X < x) = \left(\frac{x}{x_{\min}} \right)^{-(\alpha-1)} && \text{CCDF of the power law} \\
 \log P(X \geq x) &= -(\alpha - 1) \log x + (\alpha - 1) \log x_{\min} && \text{log of both sides} \\
 \log P(X \geq x) &= -0.612 \log x && \alpha = 1.612, x_{\min} = 1
 \end{aligned}$$

This is linear in $\log x$ with slope $-(\alpha - 1) = -0.612$, so a power law is a straight line on log-log axes and a bend is easy to see. The CDF itself, $1 - x^{-0.612}$, keeps the “1–”: its log does not simplify, and on log-log axes it is pressed against 1 in exactly the tail where the model and the data differ.

(g) Screen shot of the visualization of distribution of data if it was exponential

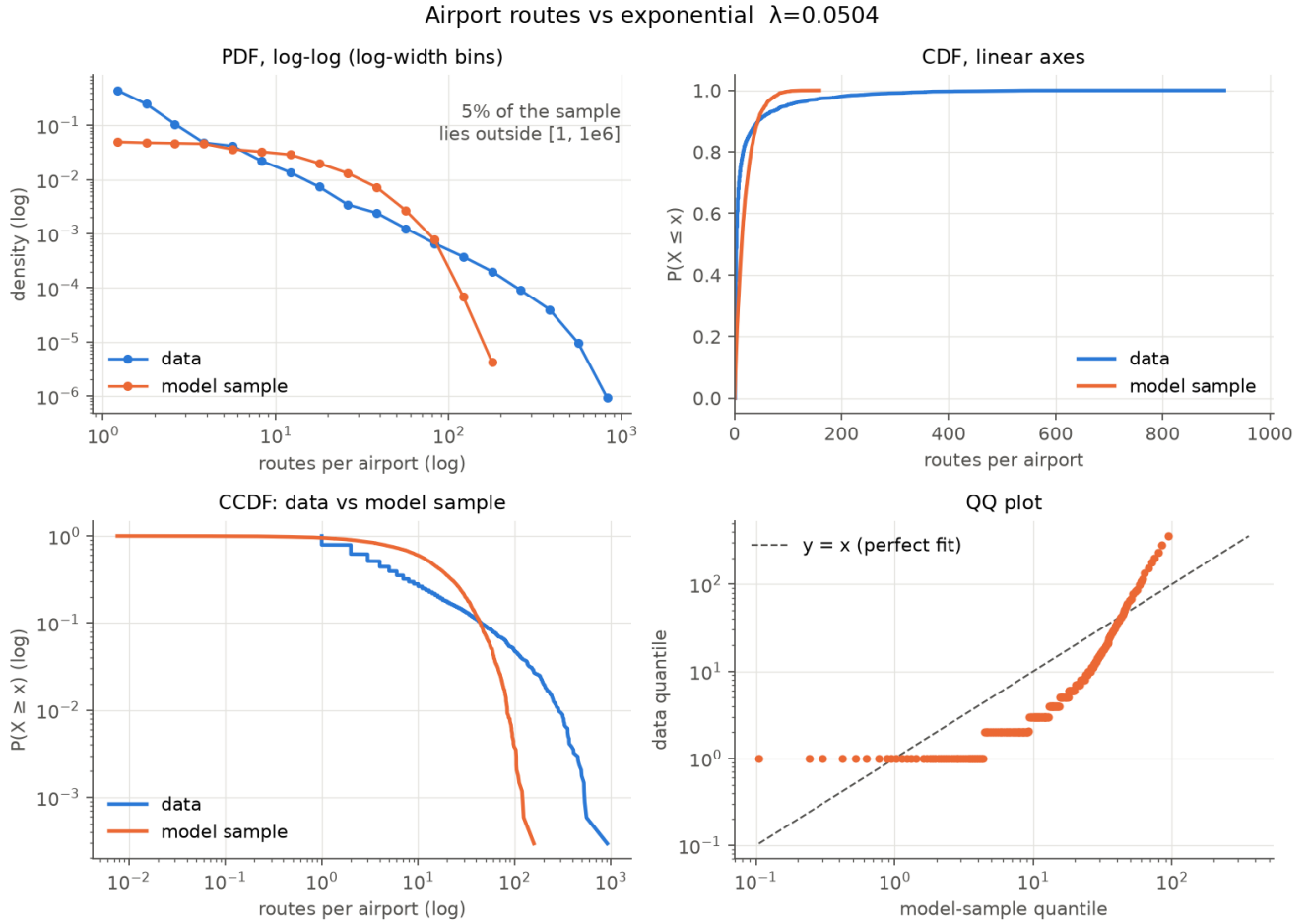


Figure 3: Airport routes vs. exponential ($\lambda = 0.0504$). $D = 0.379$.

PDF (top left). The model is almost flat for small airports and then falls off sharply after about 50 routes. The data is the opposite: steep at the start and a long, slow tail. 5% of the sample lies below 1 route, which no airport has.

CDF (top right). The model reaches 1 by about 160 routes (its largest sampled airport has 159), while the data keeps rising slowly up to 915.

CCDF (bottom left). The model bends down early and ends near 160 routes; the data goes on to 915. With $\lambda = 0.05$ an airport with 500 routes has probability about e^{-25} , yet the data has five such airports.

QQ plot (bottom right). The points are too steep: below about 40 routes they lie below $y = x$ (the model gives small airports too many routes), and above it they rise above the line (hubs get too few).

(h) Screen shot of the visualization of distribution of data if it was uniform

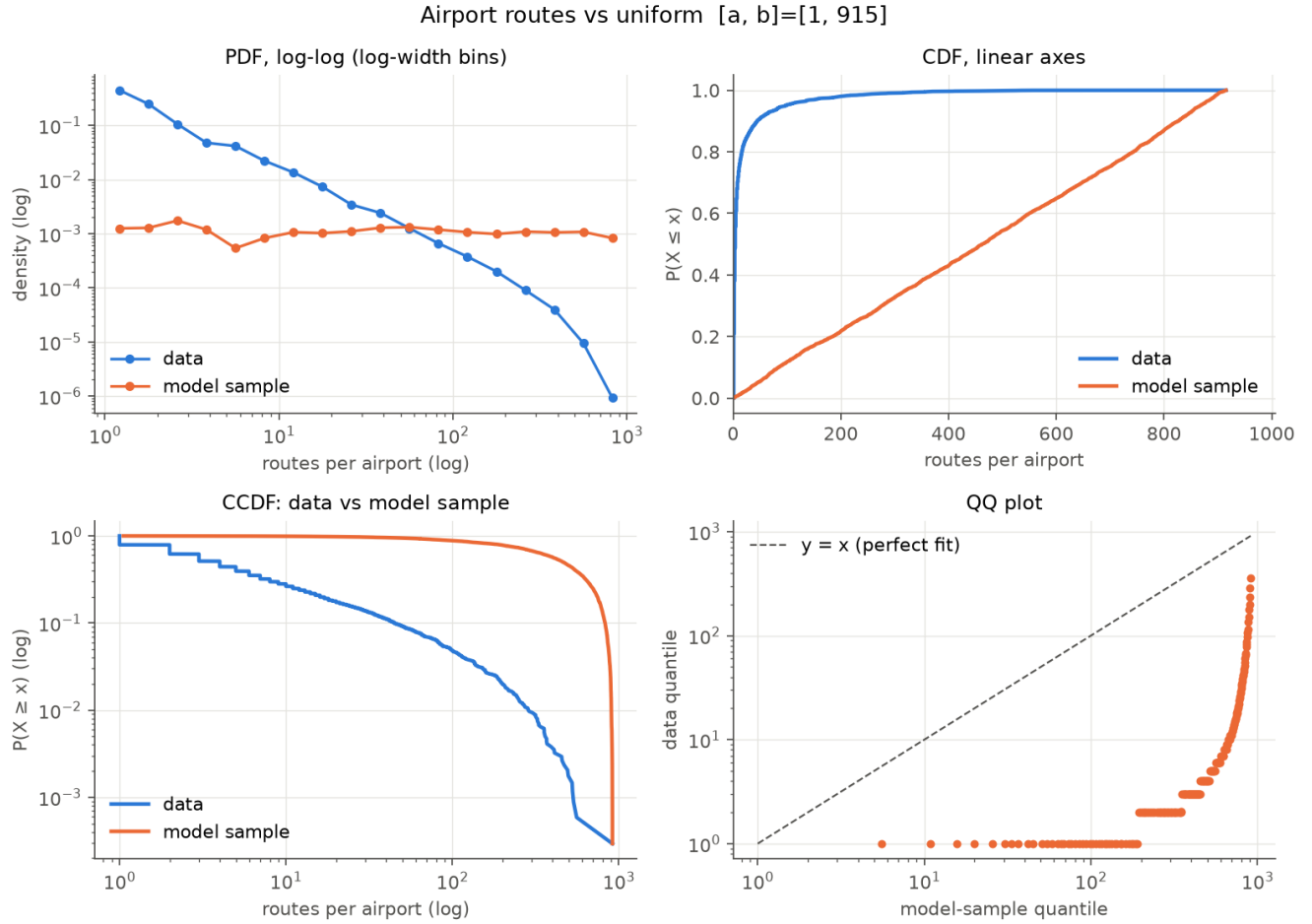


Figure 4: Airport routes vs. uniform on $[1, 915]$. $D = 0.857$.

PDF (top left). The model is a flat line near 10^{-3} across the whole range; the data falls by five orders of magnitude.

CDF (top right). The model is a straight diagonal from 1 to 915; the data reaches 0.92 by 60 routes. The largest gap between the two, $D = 0.857$, is at 60 routes.

CCDF (bottom left). The model stays near 1 until the right edge and then drops vertically at 915.

QQ plot (bottom right). Every point lies far below $y = x$, to the right. A typical sampled airport has about 460 routes, while the real median is 4 and 99% of real airports have fewer than 285.

(i) Screen shot of the visualization of distribution of data if it was normal

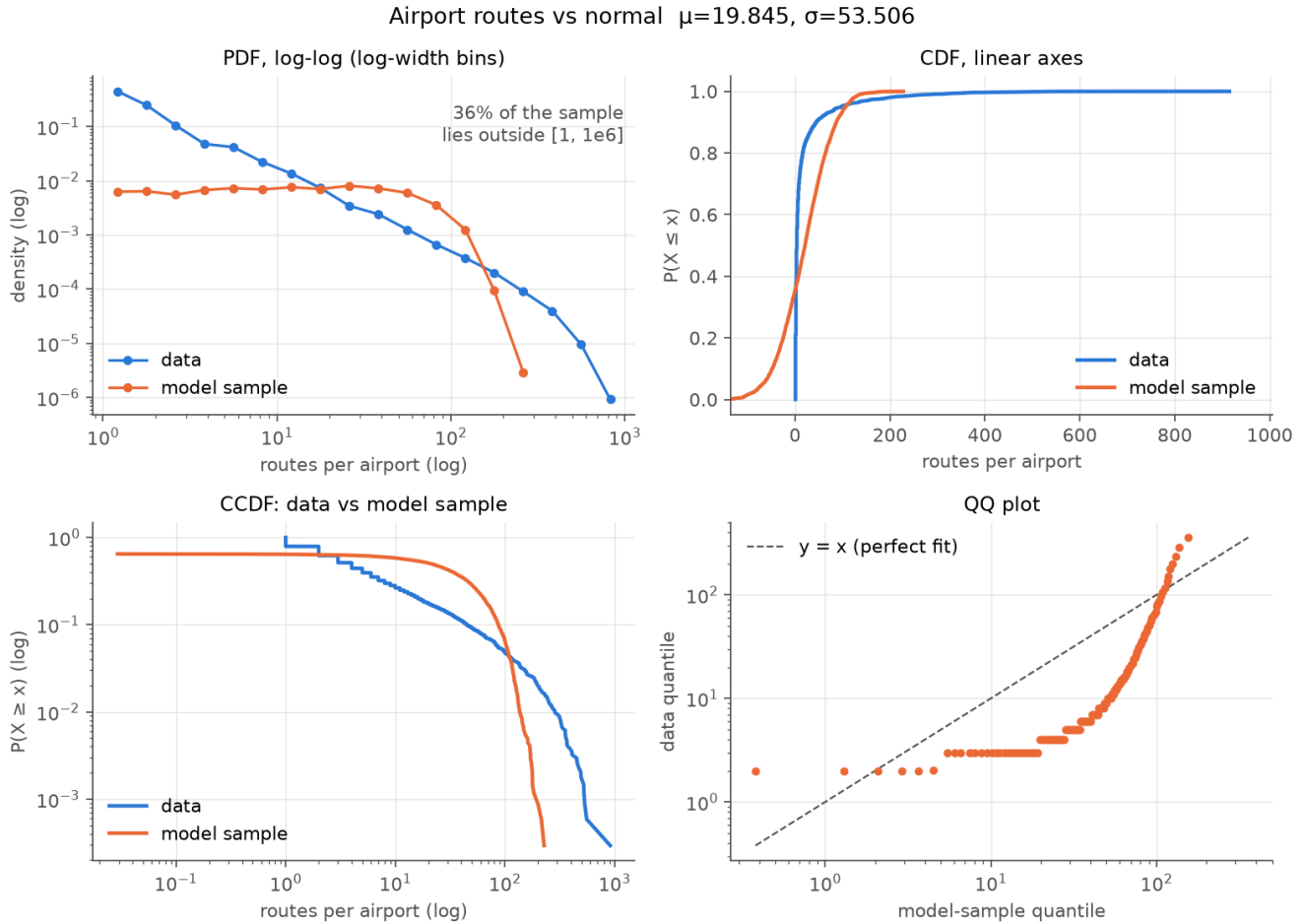


Figure 5: Airport routes vs. normal ($\mu = 19.8, \sigma = 53.5$). $D = 0.358$.

PDF (top left). The model is flat over small airports and then drops steeply after about 100 routes. 35% of its sample is below zero, which is impossible for a route count; the log axes cannot show that part.

CDF (top right). The model starts below zero (negative route counts) and is S-shaped around the mean; it reaches 1 near 230 routes while the data goes to 915.

CCDF (bottom left). Below about 100 routes the model sits above the data, and its tail ends near 230 routes, far short of 915.

QQ plot (bottom right). The points are too steep, as for the exponential: small airports get too many routes and hubs too few. The tail is thin because a normal decays like e^{-x^2} .

(j) What is the distribution of data based on your observation? Discuss

	power law	exponential	uniform	normal
KS distance D (data vs model sample)	0.209	0.379	0.857	0.358

The airport data follows a power law, with a cutoff in the far tail. It is the only one of the four models whose CCDF is a straight line on log-log axes. The data's CCDF is close to straight for more than two decades (1 to about 100 routes), and the power-law sample matches it there. The other three models fail for structural reasons, not because a parameter is slightly off:

- **Uniform** says every route count from 1 to 915 is equally likely, but most airports are tiny.
- **Normal** is symmetric and allows negative counts, while the data is bounded below at 1 and skewed to the right (skewness 6.0).
- **Exponential** has the right shape (many small, few large), but its tail falls off too fast. With $\lambda = 0.05$, an airport with 500 routes has probability about e^{-25} , yet the data has five airports with more than 500.

The power law's $D = 0.209$ comes almost entirely from $x = 1$: 20.9% of airports have exactly one route, while the continuous model puts almost no mass at or below 1. The tail mismatch above 100 routes involves few airports, so it barely moves D ; the figures show it instead.

The power law is not a perfect fit. $\hat{\alpha} = 1.61$ is below 2, so the fitted model has an infinite mean and its samples contain airports with 10^5 routes. Real airports cannot grow that large: there are only 3409 airports to fly to, and runways, gates and slots limit the busiest hubs. That is why the data's CCDF bends down above about 100 routes. The best description is a power law with an upper cutoff, which is typical of networks that grow by preferential attachment: new routes go mostly to airports that already have many routes.

Movie Rating data:

(a) What is the α if the distribution is power law?

$\alpha = 1.851$, with $x_{\min} = 1.9$ (the lowest average vote). Same MLE as for the airports, $n = 4392$.

$n = 4392$	number of movies
$x_{\min} = \min_i x_i = 1.9$	lowest average vote
$S = \sum_{i=1}^{4392} \ln(x_i/1.9) = 5163.93$	sum of log ratios
$n/S = 4392/5163.93 = 0.8505$	
$\hat{\alpha} = 1 + 0.8505 = 1.851$	estimate

(b) What is the λ if the distribution is exponential?

$\lambda = 0.1606$. MLE $\hat{\lambda} = 1/\bar{x} = 1/6.227$.

$\sum_i x_i = 27348.7$	sum of votes
$\bar{x} = 27348.7/4392 = 6.227$	mean
$\hat{\lambda} = 1/6.227 = 0.1606$	estimate

(c) What is $[a, b]$ if the distribution is uniform?

$[a, b] = [1.9, 8.5]$, the minimum and maximum votes (MLE).

$\hat{a} = \min_i x_i = 1.9$	lowest vote
$\hat{b} = \max_i x_i = 8.5$	highest vote

(d) What is μ and σ if the distribution is normal?

$\mu = 6.227$, $\sigma = 0.893$ (sample mean and sample standard deviation with $n - 1$).

$\hat{\mu} = \bar{x} = 6.227$	mean, as in the exponential fit
$\sum_i (x_i - \bar{x})^2 = 3503.28$	sum of squared deviations
$3503.28/(4392 - 1) = 0.7978$	sample variance
$\hat{\sigma} = \sqrt{0.7978} = 0.893$	standard deviation

(These can be CDF, PDF, or QQplot)

For (f) to (i):

- **Samples.** $n = 4392$ random values drawn from each fitted model (same seed).
- **Colours.** Data in blue, model sample in orange.
- **D.** Two-sample Kolmogorov–Smirnov distance: the largest vertical gap between the two CDFs (0 = identical).

(e) Screen shot of the visualization of distribution of data

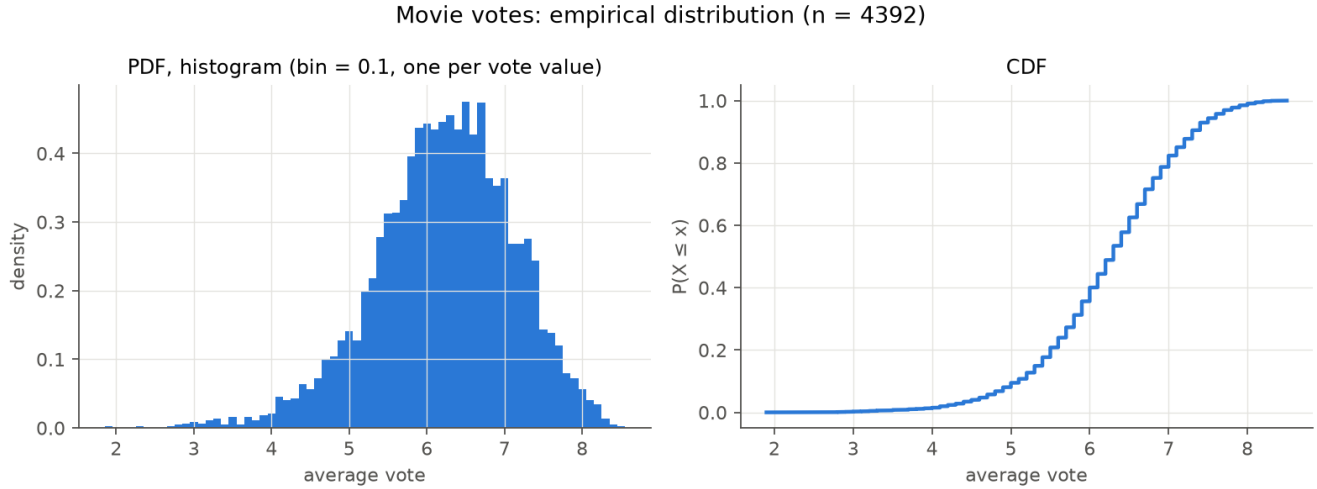


Figure 6: Movie votes: empirical distribution ($n = 4392$).

PDF (left). Histogram with one bin per possible vote (votes are rounded to 0.1). One peak near 6.3 and a nearly symmetric shape; the left tail is longer (skewness -0.48): a few movies are rated 2 to 4, and none are above 8.5.

CDF (right). An S-shaped curve centred near 6.3 (median 6.3, mean 6.23). The small steps come from the 0.1 rounding.

(f) Screen shot of the visualization of distribution of data if it was power law

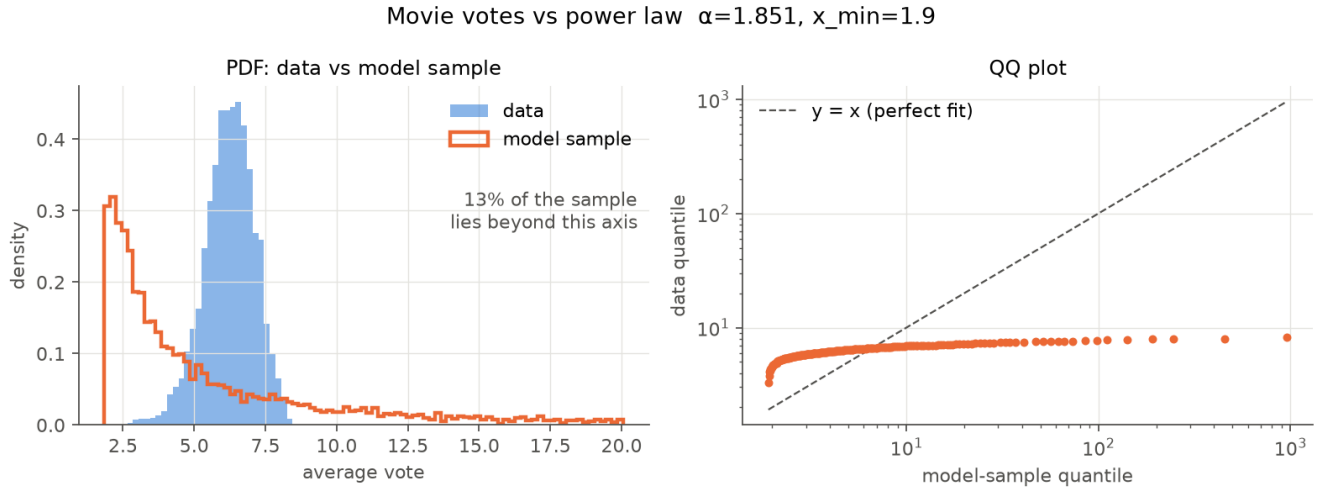


Figure 7: Movie votes vs. power law ($\alpha = 1.851$, $x_{\min} = 1.9$). $D = 0.485$.

PDF (left). The model puts its highest density at the lowest vote (1.9), where the data has almost nothing, and then a long tail. 13% of its sample lies above 20, which is not a possible rating.

QQ plot (right). Log axes, because the sample goes up to about 1000. At the top votes the points run almost horizontally to the right: the model's high quantiles are hundreds, the data's stop at 8.5.

(g) Screen shot of the visualization of distribution of data if it was exponential

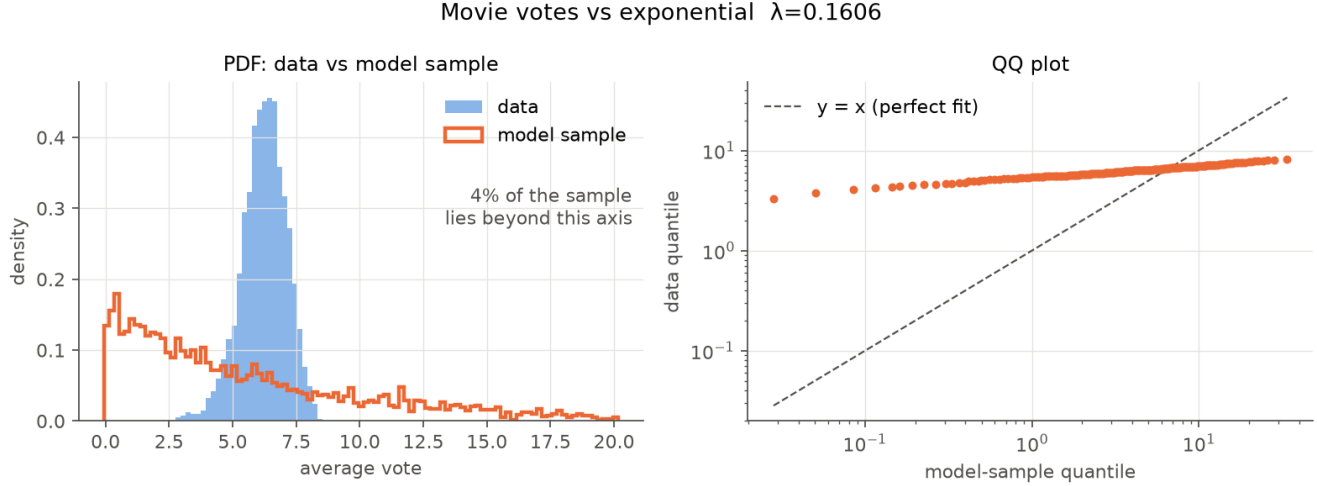


Figure 8: Movie votes vs. exponential ($\lambda = 0.1606$). $D = 0.490$.

PDF (left). The model peaks at zero and has a long right tail; 4% of its sample is above 20. The data is a narrow hump between 4 and 8.5.

QQ plot (right). The points cross $y = x$ at a shallow angle: the model's spread is far larger than the data's, with low quantiles near 0 and high quantiles above 20.

(h) Screen shot of the visualization of distribution of data if it was uniform

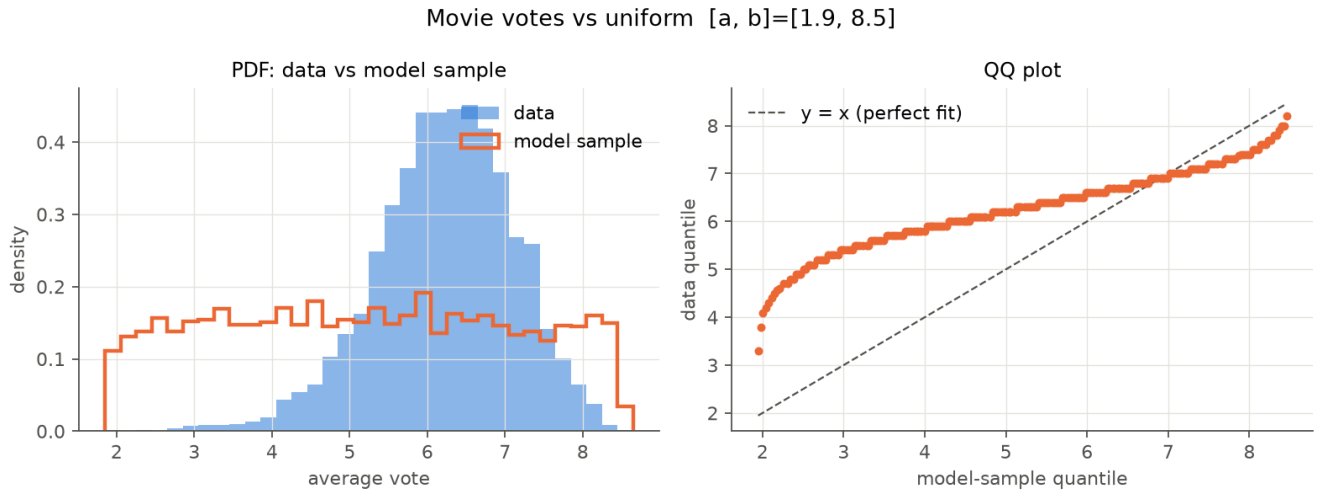


Figure 9: Movie votes vs. uniform on $[1.9, 8.5]$. $D = 0.394$.

PDF (left). The model is flat across $[1.9, 8.5]$, but the data is concentrated between 5 and 7.5.

QQ plot (right). The points lie flat across the middle, above $y = x$ on the left and below it on the right: the model's low quantiles are too low and its high quantiles too high, because it spreads the votes evenly instead of piling them in the middle.

(i) Screen shot of the visualization of distribution of data if it was normal

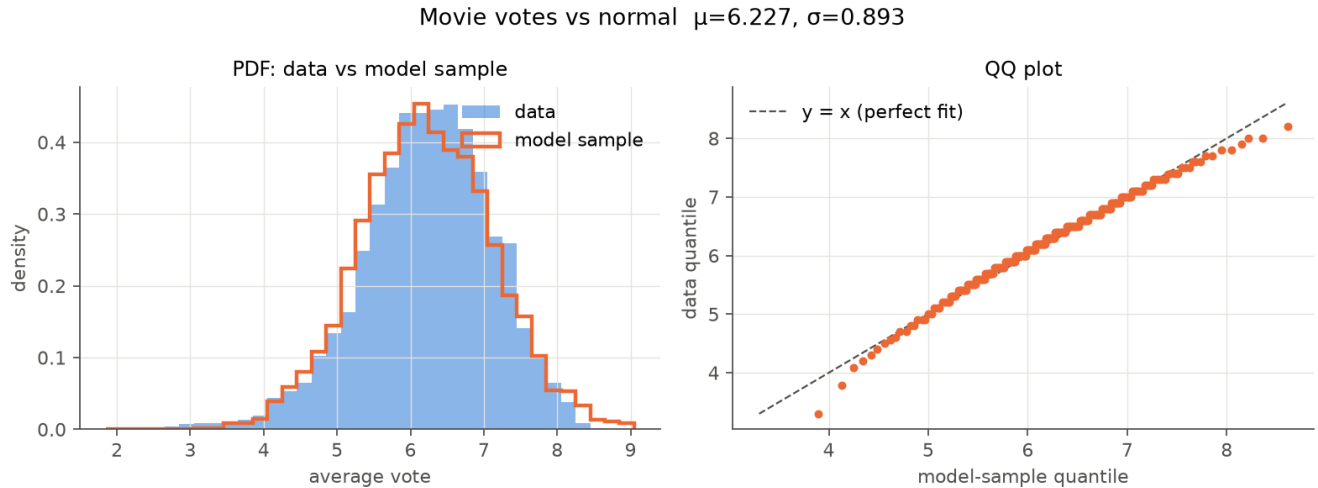


Figure 10: Movie votes vs. normal ($\mu = 6.227, \sigma = 0.893$). $D = 0.055$.

PDF (left). The sample traces the same bell as the data, with the peak in the same place.

QQ plot (right). The points lie on $y = x$ from about 4.5 to 7.5. The only departures are at the ends: the data's lowest votes are lower than the model predicts (longer left tail), and the data stops at 8.5 while the model continues. $D = 0.055$ is by far the smallest.

(j) What is the distribution of data based on your observation? Discuss

	power law	exponential	uniform	normal
KS distance D (data vs model sample)	0.485	0.490	0.394	0.055

The movie votes follow a normal distribution, approximately. The data has one peak near the middle of its range and is close to symmetric. The normal model predicts 68.3% of movies within $\pm 1\sigma$ and 95.4% within $\pm 2\sigma$; the data has 70.1% and 95.6%. In the QQ plot the points lie on the identity line through the bulk of the data, and the KS distance is about one seventh of the next-best model's.

The other three models fail for structural reasons. The power law and the exponential are both monotonically decreasing: their most likely value is the smallest one, so they predict mostly terrible movies and a long tail of ratings above 10. The uniform model ignores the peak altogether.

The fit is good but not exact, for two reasons. First, the data is bounded to $[1, 10]$ and slightly left-skewed: bad movies can drop further below the average than good movies can rise above it. Second, votes are rounded to one decimal, which gives the QQ plot its small steps. A normal shape is expected because each value is an *average* of many individual votes. By the central limit theorem an average of many independent ratings tends toward a normal distribution, whatever the distribution of each single rating.

Problem 2.

- Screenshot of distribution/histogram of the age

How the ages were computed. The birth dates have the form **dd-Mon-yy** (a few use a 4-digit year). Every 2-digit year is read as **19yy**: IMDB-WIKI was collected in 2015, so a year such as 20–24 cannot mean 2020–2024, and the counts per year decrease smoothly from the 1920s down to 00, with no break that would mark a switch to the 2000s. The 129 rows with year 0000 (a missing-value placeholder) were dropped. Per the professor’s instruction (2026-09-23), age is now the year difference **2026 minus birth year** (month/day ignored), kept only if $1 \leq \text{age} \leq 100$. Keeping that range leaves **453,905** people (mean 56.87, median 55, range 27 to 100). The minimum of 27 is a result of the **19yy** rule, since the latest possible birth year is 1999 ($2026 - 1999 = 27$). Of the rest, 6,684 rows had age > 100 and were dropped; none had age < 1 .

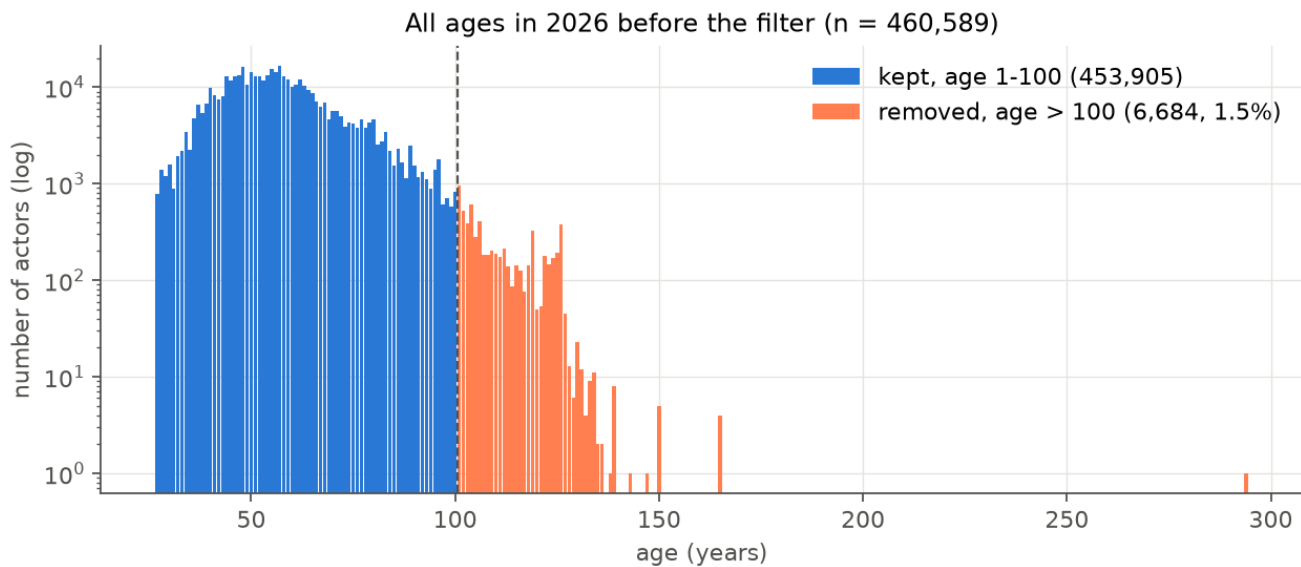


Figure 11: All ages in 2026 before the $1 \leq \text{age} \leq 100$ filter ($n = 460,589$). Blue bars are kept, coral bars are removed; log y axis.

Kept (blue). Ages 27 to 100, 453,905 people. No age is below 27 because the latest birth year is 1999.

Removed (coral). Ages 101 to 294, 6,684 people (1.5%). Most are 101 to about 130; a few isolated bars reach 150, 165 and one at 294 (born 1732). These are not plausible ages for people in a 2015 photo database, so the filter drops them. The 129 rows with year 0000 are not ages at all and are dropped before this plot.

Why log y . The removed years hold only 1.5% of people; on a linear axis their bars would be invisible next to the peak of about 16,000 per year.

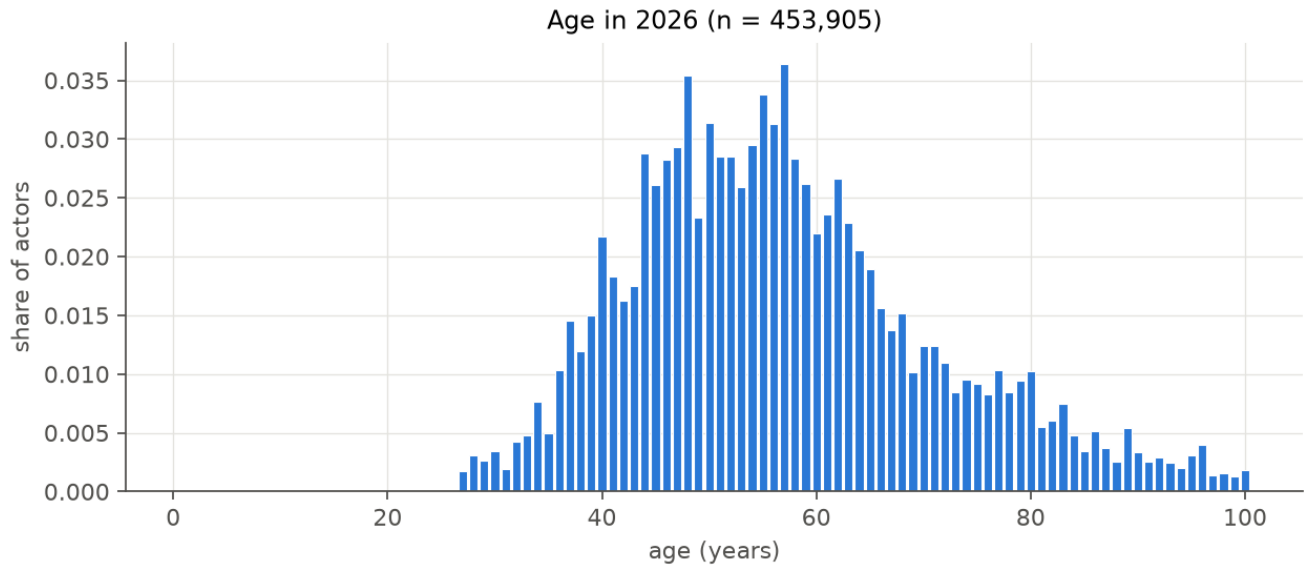


Figure 12: Age in 2026 ($n = 453,905$).

Shape. One broad hump: ages start at 27, rise through the 30s, peak in the 40s and 50s, and fall off in a long right tail up to 100. Most of the mass is between 36 and 71.

Jagged bars. Neighbouring years differ (for example, 48 and 57 stand out). This is year-to-year noise in how many actors were born in a given year, not a pattern in the ages themselves.

- Screenshot of distribution/histogram of the first digit of age

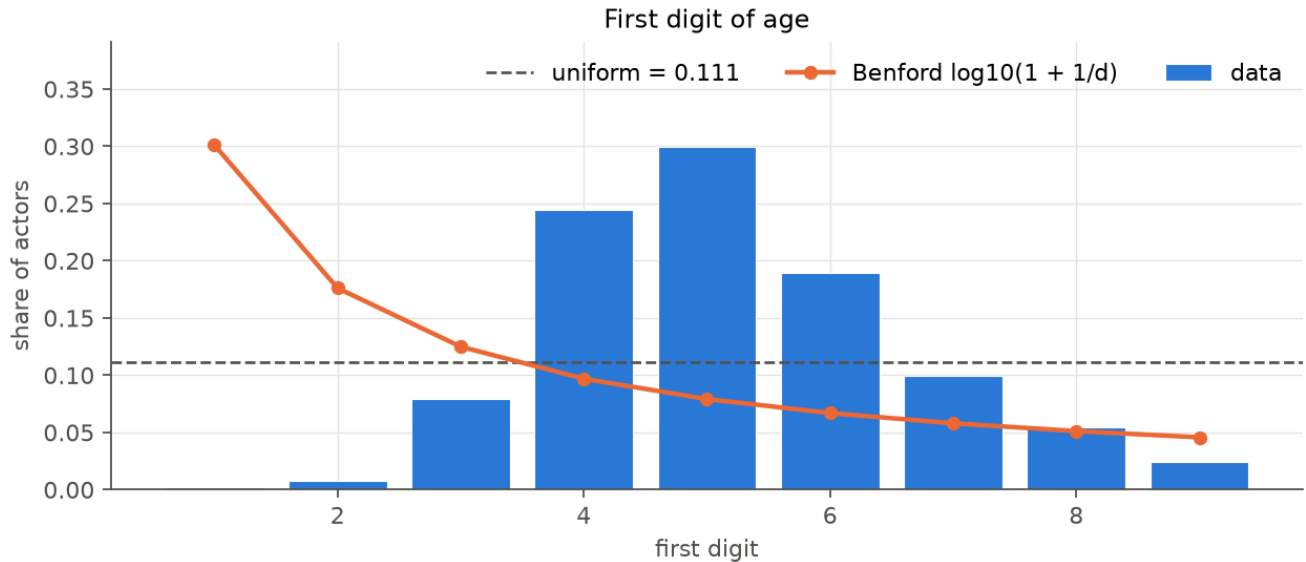


Figure 13: First digit of age, with the uniform level $1/9$ (dashed) and Benford's law (orange).

Bars. The shares are 0.2%, 0.7%, 7.9%, 24.5%, 30.0%, 18.9%, 9.9%, 5.4% and 2.5% for digits 1 to 9. They peak at 5 and 4, the same place as the hump in the age histogram.

Uniform line. Far from the dashed $1/9 = 0.111$ level: digits 4 to 6 are well above it, digits 1 to 3 and 8 to 9 well below.

Benford curve. Also far from Benford, which predicts 1 as the most common digit (30.1%); here 1 is the rarest.

- Screenshot of distribution/histogram of the last digit of age

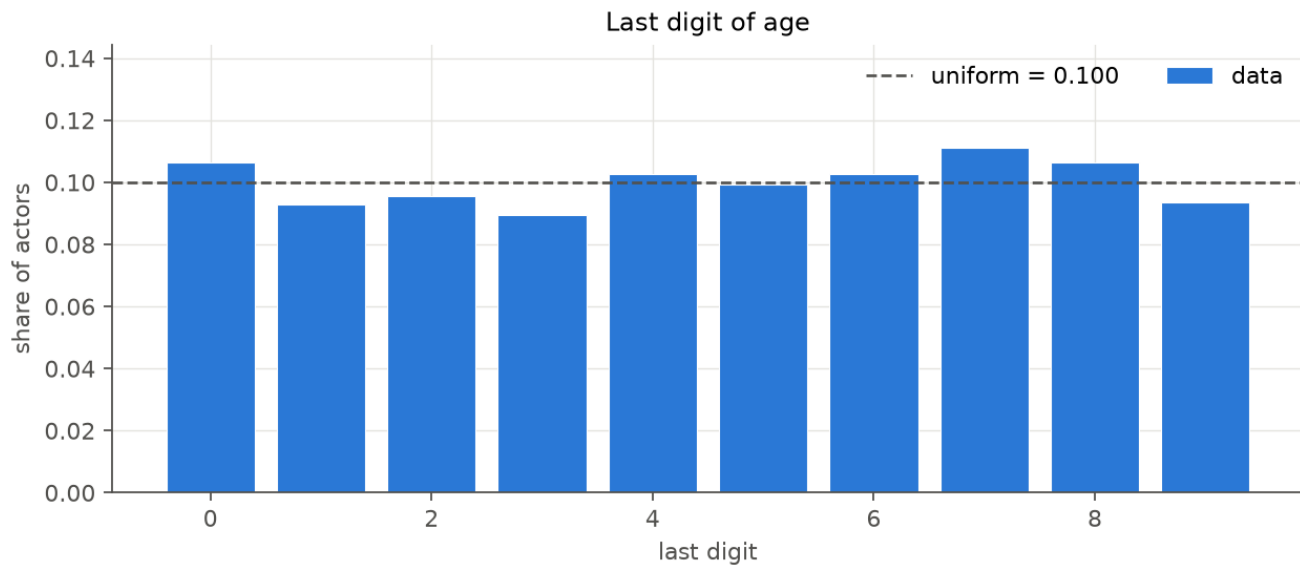


Figure 14: Last digit of age, with the uniform level 1/10 (dashed).

Bars. Every digit has a share between 8.95% (digit 3) and 11.10% (digit 7), within 1.10 percentage points of 10%.

Uniform line. All ten bars sit close to the dashed 0.100 level, with no digit clearly favoured.

- Discussion (what are those distribution?) Why do you think it is like this?

The last digit is uniform (approximately); the first digit is not.

Last digit. This is expected. The last digit of an age is its value modulo 10. Every block of ten ages (20–29, 30–39, ...) contains each last digit exactly once. The age distribution changes slowly compared with a step of one year, so within any block of ten the ten ages have roughly equal counts, and summing over the blocks gives each last digit about 10%. The remaining deviations of up to ± 1.1 points come from year-to-year noise in how many actors were born in a given year (the jagged bars in the age histogram), not from any preference for particular digits.

First digit. This is also expected, once the two boundary decades are accounted for separately. For ages 30 to 99, the first digit *is* the decade of life, and grouping by decade reproduces the shape of the age distribution: it peaks at 5 (the 50s, 30.0%) and 4 (the 40s, 24.5%), matching the median of 55. Digit 2 is small (0.7%) not because few actors are in their 20s, but because the $1 \leq \text{age} \leq 100$ rule only keeps ages 27 to 29 in that decade, three values out of ten, a boundary effect of the reference year rather than a property of the data. Digit 1 is smaller still (0.2%): no age from 10 to 19 survives the rule (the youngest possible age is 27), so the entire digit-1 count comes from the 816 people with age exactly 100 ($816/453,905 = 0.18\%$, matching the observed share). For the first digit to be uniform, every decade would need the same number of actors, which working actors clearly are not, and the two extreme digits are further skewed by where the $[1, 100]$ window happens to cut.

It is not Benford's law either. Benford's law (first digit d with probability $\log_{10}(1 + 1/d)$, so 1 is the most common at 30.1%) applies to data that spans several orders of magnitude, such as populations, prices or the airport route counts in Problem 1, where the log of the value is spread out roughly evenly. Age spans less than one order of magnitude (27 to 100) and is concentrated around 55, so its first digit shows where that one hump sits, not Benford's pattern. In short, the last digit is uniform because it throws away where

the value is and keeps only its position within a block of ten. The first digit keeps exactly that information, so its distribution follows the shape of the age distribution.

Tooling. Python 3 / pandas / numpy / scipy / matplotlib; all estimates computed programmatically, none typed by hand; analysis code, figure code and write-up drafted with Claude Code. Code: one script per answer in `hw2/code/` (for example `p1_airport_a_alpha.py`), sharing `common.py` and `common_p2.py`; `run_all.py` runs them all.